

4.18.3.1. **MsgCounterSyncReq** - Message Counter Synchronization Request

The Message Counter Synchronization Request (*MsgCounterSyncReq*) message is sent when a message was received from a peer whose current message counter is unknown.

A *MsgCounterSyncReq* message SHALL set the **C Flag** in the message header. The control message counter SHALL be used for message protection.

A *MsgCounterSyncReq* message SHALL be secured with the group key for which counter synchronization is requested and SHALL set the **Session Type** to **1**, indicating a group session as per the rules outline in [Section 4.18.5, “Message Counter Synchronization Exchange”](#).

The payload of the *MsgCounterSyncReq* message takes the format defined in [Table 26, “Message Counter Sync Request”](#):

Table 26. Message Counter Sync Request

Field Size	Message Field	Description
8 bytes	Challenge	The Challenge is a 64-bit random number generated using the DRBG by the initiator of a <i>MsgCounterSyncReq</i> to uniquely identify the synchronization request cryptographically.

4.18.3.2. **MsgCounterSyncRsp** - Message Counter Synchronization Response

The Message Counter Synchronization Response (*MsgCounterSyncRsp*) message is sent in response to a *MsgCounterSyncReq*.

A *MsgCounterSyncRsp* message SHALL set the **C Flag** in the message header. The control message counter SHALL be used for message protection.

The *MsgCounterSyncRsp* message has the format defined in [Table 27, “Message Counter Sync Response”](#):

Table 27. Message Counter Sync Response

Field Size	Message Field	Description
4 bytes	Synchronized Counter	The current data message counter for the node sending the <i>MsgCounterSyncRsp</i> message.
8 bytes	Response	The Response SHALL be the same as the 64-bit value sent in the Challenge field of the corresponding <i>MsgCounterSyncReq</i> .

4.18.4. Unsynchronized Message Processing

An unsynchronized message is one that is cryptographically verified from a node whose message counter is unknown. Upon receipt of an unsynchronized message process the message as follows: